

AI-Powered Multi-Agent Research Assistant using LangGraph

Use Case	Skills Covered	Expected Outcome	Complexity	Industry
Build an AI-Powered Multi-Agent Research Assistant	LangChain, LangGraph, StateGraph, Multi-Agent Systems, Supervisor Pattern, Tool Calling, CrewAI, Agentic Memory, Human-in-the-Loop (HITL), Text-to-SQL/MYSQL, LangSmith, Pydantic	Design and implement a production-grade multi-agent AI system capable of autonomous reasoning, web research, structured database querying, persistent memory, compliance approval, and report generation	Medium-Complex	Market Research / Enterprise AI

Problem Statement

Market intelligence firms continuously analyze competitors, products, pricing, industry trends, and internal business data to prepare executive reports. Traditional research workflows require multiple analysts to manually gather information from the web, query enterprise databases, verify facts, and prepare structured reports before obtaining managerial approval.

Organizations require an intelligent **Multi-Agent AI System** capable of autonomously coordinating multiple specialized agents to perform these tasks while maintaining conversation history, validating generated outputs, and supporting human approvals.

Your task is to build a complete **Multi-Agent Research Assistant** using **LangChain** and **LangGraph** that employs the **Supervisor Multi-Agent Pattern**. The system should intelligently route tasks between specialized agents, utilize external tools, maintain persistent memory, query structured enterprise data, generate professional research reports, and support Human-in-the-Loop approval before delivering the final output.

The final solution should support:

- Autonomous task planning using a Supervisor Agent
- Intelligent routing between multiple specialized agents
- Web research using external search tools
- Internal business data retrieval using Text-to-SQL/MYSQL
- Persistent episodic and semantic memory
- Human approval workflow before report generation
- Streaming responses through CLI or web application

- End-to-end observability using LangSmith

Dataset / Resources

Knowledge Base

- Industry News Articles
- Competitor Websites
- Product Documentation
- Market Research Reports

Structured Data

- sales.db (SQL/MYSQLite Database)
- products.csv
- competitors.csv
- quarterly_sales.csv

Tools to Implement

1. **Web Research Tool**
 - a. Search market trends and competitor information
2. **Text-to-SQL/MYSQL Tool**
 - a. Convert natural language queries into SQL/MYSQL
 - b. Validate SQL/MYSQL before execution
 - c. Query SQL/MYSQLite database
3. **Report Writer Tool**
 - a. Generate executive-ready competitor reports
4. **Memory Tool**
 - a. Store and retrieve conversation history using Redis/Zep
5. **Human Approval Tool**
 - a. Route reports to compliance or manager approval before final delivery

Instructions for Milestone - 3

- Design a **LangGraph StateGraph** with typed graph states and conditional routing between agents.
- Build a **Supervisor Agent** responsible for planning tasks and delegating work to specialized agents.
- Develop a **Researcher Agent** that performs web searches and retrieves structured enterprise data using a Text-to-SQL workflow.
- Implement a **Writer Agent** that compiles research findings into a professional competitor analysis report.

- Integrate **persistent memory (Redis/Zep)** to maintain both episodic and semantic conversation history across sessions.
- Add a **Human-in-the-Loop (HITL)** approval gate that reviews generated reports before final submission.
- Implement a **Text-to-SQL validation loop** to verify SQL queries before execution against the SQLite database.
- Use **Pydantic** models to define and validate all tool input/output schemas.
- Enable **structured JSON logging** for agent execution, tool calls, routing decisions, and system events.
- Stream intermediate and final outputs through a CLI or FastAPI/Streamlit interface.
- Trace the complete workflow using **LangSmith** for debugging and observability.
- Create a comprehensive **pytest** suite to validate agent routing, memory recall, tool execution, SQL validation, and report generation using at least **20 synthetic test scenarios**.
- Demonstrate routing accuracy, memory recall performance, and successful multi-agent collaboration through sample executions.